

DIGITAL LOGIC DESIGN

CH-1: NUMBER REPRESENTATION AND CODES

DECIMAL	BINARY	HEXADECIMAL	OCTAL
0	0000	0	000
1	0001	1	001
2	0010	2	010
3	0011	3	011
4	0100	4	100
5	0101	5	101
6	0110	6	110
7	0111	7	111
8	1000	8	
9	1001	9	BCD
10	1010	A	DIRECT CONVERSIONS
11	1011	B	Eg: 15 - <u>00010101</u>
12	1100	C	not 1111
13	1101	D	
14	1110	E	
15	1111	F	

CONVERSIONS:

DECIMAL $\rightarrow$ BINARY : $(NO.)_{10} \div 2 \rightarrow (NO.)_2$
 $\rightarrow$ OCTAL : $(NO.)_{10} \div 8 \rightarrow (NO.)_8$
 $\rightarrow$ HEXA : $(NO.)_{10} \div 16 \rightarrow (NO.)_{16}$ } ADD REMAINDERS
TO DIGIT POSITION

BINARY $\rightarrow$ DECIMAL : $(NO.)_2 \times 2^{\text{position}}$: $(1111)_2 \rightarrow 1 \times 2^0 + 1 \times 2^1 \dots = 15$
 $\rightarrow$ OCTAL : Group no. into 3-bit groups & directly convert
 $\rightarrow$ HEXA : Group no. into 4-bit groups & directly convert

OCTAL $\rightarrow$ DECIMAL: $(NO.)_8 \times 8^{\text{position}}$
 $\rightarrow$ BINARY: Directly convert each digit to 3-digit binary
 $\rightarrow$ HEXA: OCTAL $\rightarrow$ BINARY $\rightarrow$ HEXA

HEXA $\rightarrow$ DECIMAL: $(NO.)_{16} \times 16^{\text{position}}$
 $\rightarrow$ BINARY: Directly convert each digit to 4-bit binary
 $\rightarrow$ OCTAL: HEXA $\rightarrow$ BINARY $\rightarrow$ OCTAL

QUESTIONS:

1. $(157)_8 \rightarrow ()_2$

$\Rightarrow 1-001, 5-101, 7-111$
 $\Rightarrow (00110111)_2$

2. $(2C)_{16} \rightarrow ()_{10}$

$\Rightarrow (2 \times 16^1) + (C \times 16^0)$
 $\Rightarrow 32 + 12 \Rightarrow (44)_{10}$

3. $(245)_{10} \rightarrow ()_8, ()_{16}$

$\Rightarrow \begin{array}{r|l} 8 & 245 \\ 8 & 30 \rightarrow 5 \\ 8 & 3 \rightarrow 6 \\ & 0 \rightarrow 3 \end{array} \Rightarrow (365)_8$

$\Rightarrow \begin{array}{r|l} 16 & 245 \\ 16 & 5 \rightarrow 5 \\ & 0 \rightarrow 15 \end{array} \Rightarrow (F5)_{16}$

4. $(101101)_2 \rightarrow ()_{16}$

$\Rightarrow \begin{array}{cc} \underline{0010} & \underline{1101} \\ 2 & 13 \end{array} \Rightarrow (2D)_{16}$

5. $(111010)_2 \rightarrow \text{BCD}$

$$\Rightarrow 0 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 1 \times 2^5$$

$$2 + 8 + 16 + 32$$

$$(58)_{10} \rightarrow 5 - 0101$$

$$8 - 1000$$

$$\Rightarrow (01011000)$$

BINARY ADDITION:

Eg: $12 + 7 :$

$$\begin{array}{r} 1100 \\ + 0111 \\ \hline 10011 \end{array}$$

$\Rightarrow (19)_{10}$

COMPLEMENTS:

1's: $0 \leftrightarrow 1$

2's: $0 \leftrightarrow 1, + 1$

BINARY SUBTRACTION:

1. $12 - 7 :$

$$\begin{array}{r} 12 - 01100 \\ 7 - 00111 \\ \hline 1's : 7 - 11000 \end{array}$$

$$\begin{array}{r} 101100 \\ + 11000 \\ \hline (1) 00100 \end{array}$$

$\Rightarrow 00100 \rightarrow 4 + 1 = \underline{5}$

by 1's

add remainder

2. $7 - 12 :$

$$\begin{array}{r} 7 - 00111 \\ 12 - 01100 \\ \hline 1's : 12 - 10011 \end{array}$$

$$\begin{array}{r}
 00111 \\
 + 10011 \\
 \hline
 11010
 \end{array}$$

$$1's: 00101 \rightarrow \underline{-5}$$

↳ no remainder, take 1's then add a '-'

3. 12 - 7:

$$12: 01100$$

$$7: 00111$$

$$2's: 7 - 11000 = 11001$$

$$\begin{array}{r}
 + 1 \\
 \hline
 11001
 \end{array}$$

$$\Rightarrow 101100$$

$$+ 11001$$

ignore ← (1) 00101

$$\Rightarrow 00101 = \underline{5}$$

4. 7 - 12:

$$7: 00111$$

$$12: 01100$$

$$2's: 12: 10111 = 10100$$

$$\begin{array}{r}
 + 1 \\
 \hline
 10100
 \end{array}$$

$$\Rightarrow 00111$$

$$+ 10100$$

$$11011$$

$$2's: 00100$$

$$+ 1$$

$$00101 = \underline{-5}$$

→ no remainder, so take 2's and add '-'

GRAY CODE:

DECIMAL

BINARY

GRAY

0	0000	0000
1	0001	0001
2	0010	0011
3	0011	0010
4	0100	0110
5	0101	0111
6	0110	0101
7	0111	0100

BINARY TO GRAY CODE :

BINARY	b ₃	b ₂	b ₁	b ₀
	1	1	0	0
	1			
g ₃	1			
g ₃ g ₂	1	0		
g ₃ g ₂ g ₁	1	0	1	
g ₃ g ₂ g ₁ g ₀	1	0	1	0

GRAY CODE TO BINARY :

Same as binary to gray.

PARITY FOR ERROR DETECTION:

Even parity: even no. of 1's $\rightarrow$ 0 is added
 odd no. of 1's $\rightarrow$ 1 is added

Odd parity: even no. of 1's $\rightarrow$ 1 is added
 odd no. of 1's $\rightarrow$ 0 is added

Eg: 1101

even: parity bit = 1

odd: parity bit = 0

CH-2: LOGIC REPRESENTATION AND MINIMISATION TECHNIQUES

LOGIC GATES:

NAME	DIAGRAM	TRUTH TABLE
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AND		
-----	---	--

If both are 1, output $\rightarrow$ 1, otherwise $\rightarrow$ 0

$$X = A \cdot B$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR		
----	---	--

If both are 0, output $\rightarrow$ 0, otherwise $\rightarrow$ 1

$$X = A + B$$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT		
-----	---	--

Returns opp. of i/p as o/p

$$X = A'$$

A	A' / $\bar{A}$
0	1
1	0

NAND		
------	---	--

If both are 1 $\rightarrow$ 0, otherwise $\rightarrow$ 1

$$X = (A \cdot B)'$$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

$\rightarrow$ UNIVERSAL GATE

NOR



A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

If both are 0 $\rightarrow$ 1, otherwise $\rightarrow$ 0

$$X = (A + B)'$$

$\rightarrow$ UNIVERSAL GATE

XOR



A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

If both are same, o/p $\rightarrow$ 0, otherwise $\rightarrow$ 1

$$X = A \oplus B$$

BOOLEAN ALGEBRA:

1. IDENTITY : $A + 0 = A$ $A \cdot 1 = A$
2. NULL : $A + 1 = 1$ $A \cdot 0 = 0$
3. IDEMPOTENT: $A + A = A$ $A \cdot A = A$
4. COMPLEMENT: $A + A' = 1$ $A \cdot A' = 0$
5. DOUBLE NEGATION LAWS: $(A')' = A$
6. COMMUTATIVE : $A + B = B + A$ $A \cdot B = B \cdot A$
7. ASSOCIATIVE : $(A + B) + C = A + (B + C)$ $(A \cdot B) \cdot C = A \cdot (B \cdot C)$
8. DISTRIBUTIVE: $A \cdot (B + C) = A \cdot B + A \cdot C$ $A + (B \cdot C) = (A + B) \cdot (A + C)$
9. ABSORPTION: $A + A \cdot B = A$ $A \cdot (A + B) = A$
10. DEMORGAN'S: $(A \cdot B)' = A' + B'$ $(A + B)' = A' \cdot B'$
11. REDUNDANCY: $A + A' \cdot B = A + B$ $A \cdot (A' + B) = A \cdot B$

SIMPLIFICATION:

Q.1.

- NO. OF GATES: AND $\rightarrow$ no. of terms in a term - 1 (for each term)
 OR $\rightarrow$ total no. of terms - 1
 NOT $\rightarrow$ no. of complements

Eg : $Y = AB + AB'C + C' + ABC$

AND = $(2-1) + (3-1) + (1-1) + (3-1) = 1+2+0+2 = 5$

OR = $4-1 = 3$

NOT = 2

→ 2-input gates

SIMPLIFICATION: $Y = AB + AB'C + C' + ABC$

$$\begin{aligned}
 &\Rightarrow AB + ABC + AB'C + C' \\
 &\quad AB(1+C) + AB'C + C' \\
 &= AB + AB'C + C' \\
 &= A(B + B'C) + C' \\
 &= A(B + C) + C' \\
 &= AB + AC + C' \\
 &= AB + A + C' \\
 &= A(B+1) + C' = A + C'
 \end{aligned}$$

$\therefore Y = A + C'$

1 NOT GATE

1 NOR GATE

TRUTH TABLE:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Q.2. $F(A, B, C, D) = (A+B') \cdot (ACB + B' \cdot [C+A]) + ABC' \cdot (A'B + A)$

D is not required

$$\Rightarrow (A+B') \cdot (ABC + B'C + B'A) + ABC' \cdot (A+B)$$

$$(A+B') \cdot (ABC + B'C + B'A) + ABC'A + ABC'B$$

$$(A+B') \cdot (ABC + B'C + B'A) + ABC'$$

$$\Rightarrow ABC + AB'C + AB' + B'C + B'A + ABC'$$

$$ABC + AB'(1+C) + B'(C+A) + ABC'$$

$$ABC + AB' + B'C + ABC'$$

$$ABC(C+C') + AB' + B'C$$

$$AB + AB' + B'C$$

$$A(B+B') + B'C$$

$$A + B'C$$

$$\therefore \text{AND} = 1$$

$$\text{OR} = 1$$

$$\text{NOT} = 1$$

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

SOP AND POS:

 **SOP**
MINTERMS: represents logic 1. Product of all input variables such that only 1 combination gives 1.

 **POS**
MAXTERMS: represents logic 0. Sum of all input variables such that only one combination results in 0.

Eg: SOP : $A'B'CD + A'B'CD + AB'CD$

POS : $(A+B+C+D) \cdot (A+B+C+D') \cdot (A+B'+C'+D) \cdot (A'+B'+C+D) \cdot (A'+B'+C'+D')$

SOP - OR of AND'S ; AND-OR networks

POS - AND of OR'S ; OR-AND networks

Q.1. Truth table for F_{4m} which is 1 when 4-digit number is prime.

A	B	C	D	F
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

→ write 1 for all prime numbers

minterms : $F(A, B, C, D) = \sum m(2, 3, 5, 7, 11, 13)$

maxterms : $F(A, B, C, D) = \prod M(0, 1, 4, 6, 8, 9, 10, 12, 14, 15)$

→ write all minterms to convert to a Boolean expression

$P = A'B'CD' + A'B'CD + A'BC'D + A'BCD + AB'CD + ABC'D$

$$\Rightarrow A'B'C(CD' + D) + A'BD(C' + C) + AB'CD + ABC'D$$

$$A'B'C + A'BD + AB'CD + ABC'D$$

Q.2. $Y(A, B, C, D) = \sum m(2, 3, 5, 7, 8, 10, 11, 13) \rightarrow \text{SOP}$

2	-	0010	-	$A'B'CD'$
3	-	0011	-	$A'B'CD$
5	-	0101	-	$A'BC'D$
7	-	0111	-	$A'BCD$
8	-	1000	-	$AB'C'D'$
10	-	1010	-	$AB'CD'$
11	-	1011	-	$AB'CD$
13	-	1101	-	$ABC'D$

$$Y = A'B'CD' + A'B'CD + A'BC'D + A'BCD + AB'C'D' + AB'CD' + AB'CD + ABC'D$$

$$= A'B'C + A'BD + AB'D' + AB'CD + ABC'D$$

$$= A'B'C + A'BD + AB'(D' + CD) + ABC'D$$

$$= A'B'C + BD(A' + AC') + AB'(CD' + CD)$$

$$= A'B'C + BD(A' + C') + AB'(CD' + C)$$

$$= A'B'C + A'BD + BC'D + AB'D' + AB'C$$

$$= B'C(A' + A) + A'BD + BC'D + AB'D'$$

$$= B'C + A'BD + BC'D + AB'D'$$

$$\therefore Y = B'C + A'BD + BC'D + AB'D'$$

KARNAUGH MAPS:

Q.1. $Y(A, B, C) = \sum m(1, 2, 3, 4, 7) \quad (\text{SOP})$

		C	
AB		0	1
00		0 ₀	1 ₁
01		1 ₂	1 ₃
11		0 ₆	1 ₇
10		1 ₄	0 ₅

$$(m_1, m_3) \rightarrow A'C$$

$$(m_2, m_3) \rightarrow A'B$$

$$(m_3, m_7) \rightarrow BC$$

$$(m_4) \rightarrow AB'C'$$

$$Y = A'C + A'B + BC + AB'C'$$

Q.2. $Y(A, B, C, D) = \sum m(1, 2, 3, 4, 7, 8, 11, 13, 15)$

AB \ CD	00	01	11	10
00	0 ₀	1 ₁	1 ₂	1 ₃
01	1 ₄	0 ₅	1 ₆	0 ₇
11	0 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
10	1 ₈	0 ₉	1 ₁₁	0 ₁₀

$$(m_1, m_3) \rightarrow A'B'D$$

$$(m_3, m_2) \rightarrow A'B'C$$

$$(m_3, m_7, m_{15}, m_{11}) \rightarrow CD$$

$$(m_{13}, m_{15}) \rightarrow ABD$$

$$(m_4) \rightarrow A'BC'D'$$

$$(m_8) \rightarrow AB'C'D'$$

$$\therefore Y = A'B'D + A'B'C + CD + ABD + A'BC'D' + AB'C'D'$$

Q.3. $Y(A, B, C, D) = \sum m(0, 1, 2, 3, 4, 6, 7, 8, 11, 13, 15)$

AB \ CD	00	01	11	10
00	1 ₀	1 ₁	1 ₂	1 ₃
01	1 ₄	0 ₅	1 ₆	1 ₇
11	0 ₁₂	1 ₁₃	1 ₁₅	0 ₁₄
10	1 ₈	0 ₉	1 ₁₁	0 ₁₀

$$(m_0, m_1, m_2, m_3) \rightarrow A'B'$$

$$(m_0, m_4, m_2, m_6) \rightarrow A'D'$$

$$(m_{13}, m_{15}) \rightarrow ABD$$

$$(m_3, m_7, m_{15}, m_{11}) \rightarrow CD$$

$$(m_0, m_8) \rightarrow B'C'D'$$

$$\therefore Y = A'B' + CD + A'D' + B'C'D' + ABD$$

Q.4. $Y(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 11, 15) + d(3, 9, 13)$

AB \ CD	00	01	11	10
00	1 ₀	1 ₁	X ₃	1 ₂
01	1 ₄	0 ₅	0 ₇	1 ₆
11	0 ₁₂	X ₈	1 ₁₅	0 ₁₄
10	0 ₈	X ₉	1 ₁₁	0 ₁₀

$$(m_0, m_1, m_3, m_2) \rightarrow A'B'$$

$$(m_0, m_4, m_2, m_6) \rightarrow A'D' \quad (m_{13}, m_{15}, m_9, m_{11}) \rightarrow AD$$

$$\therefore Y = A'B' + A'D' + AD$$

Q.5. $Y(A, B, C, D) = \prod M(5, 9, 10, 12, 14)$

AB \ CD	00	01	11	10
00	1 ₀	1 ₁	1 ₃	1 ₂
01	1 ₄	0 ₅	1 ₇	1 ₆
11	0 ₁₂	1 ₈	1 ₁₅	0 ₁₄
10	1 ₈	0 ₉	1 ₁₁	0 ₁₀

$$(m_{14}, m_{10}) \rightarrow (A' + C' + D) \quad (m_5) \rightarrow (A + B' + C + D')$$

$$(m_{12}, m_{14}) \rightarrow (A' + B' + D) \quad (m_9) \rightarrow (A' + B + C + D')$$

$$\therefore Y = (A' + C' + D) \cdot (A' + B' + D) \cdot (A + B' + C + D') \cdot (A' + B + C + D')$$

Q.6. $F(A, B, C, D) = \sum m(0, 2, 4, 5, 8, 10, 12, 14, 15)$

AB \ CD	00	01	11	10
00	1 ₀	0 ₁	0 ₃	1 ₂
01	1 ₄	1 ₅	0 ₇	0 ₆
11	1 ₁₂	0 ₁₃	1 ₁₅	1 ₁₄
10	1 ₈	0 ₉	0 ₁₁	1 ₁₀

$$(m_0, m_4, m_{12}, m_8) \rightarrow C'D' \quad (m_0, m_8, m_2, m_{10}) \rightarrow B'D'$$

$$(m_4, m_5) \rightarrow A'BC'$$

$$(m_{15}, m_{14}) \rightarrow ABC$$

$$(m_{12}, m_8, m_{14}, m_{10}) \rightarrow AD'$$

$$\therefore Y = C'D' + B'D' + AD' + A'BC' + ABC$$

NAND AS A UNIVERSAL GATE:

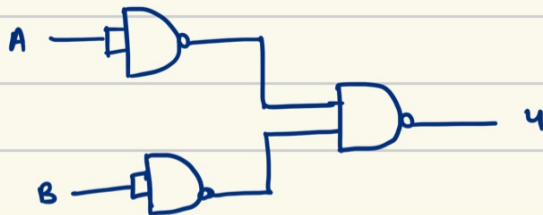
NOT :



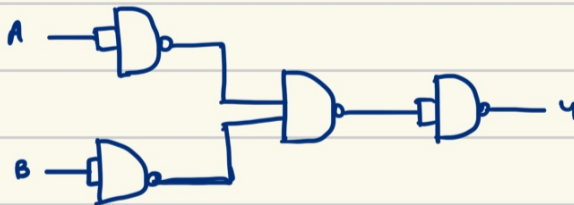
AND :



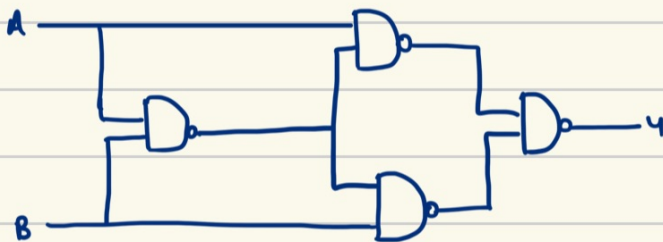
OR :



NOR :



XOR :

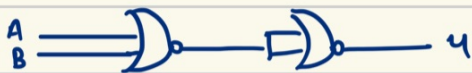


NOR AS A UNIVERSAL GATE:

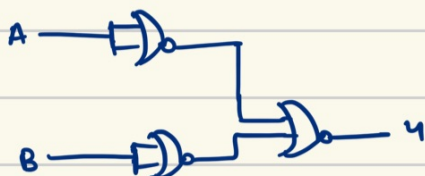
NOT :



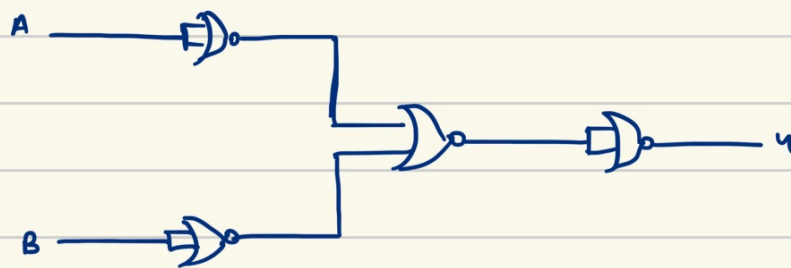
OR :



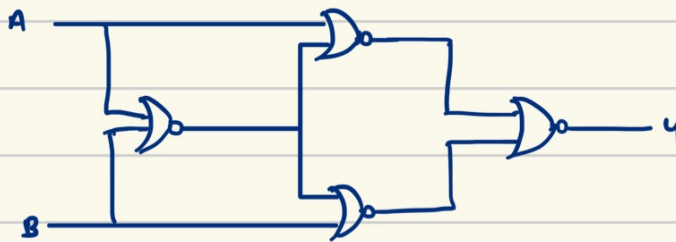
AND :



NAND :



EXNOR :



NUMERICALS:

1. $F(A, B, C) = \sum m(3, 4, 6, 7)$

AB \ C	0	1
00	0	0
01	0	1
11	1	1
10	1	0

$(m_6, m_4) \rightarrow AC'$

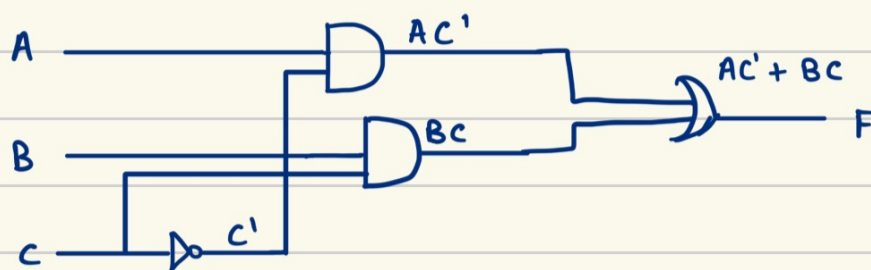
$(m_3, m_7) \rightarrow BC$

$\therefore F = AC' + BC$

2 AND

1 NOT

1 OR



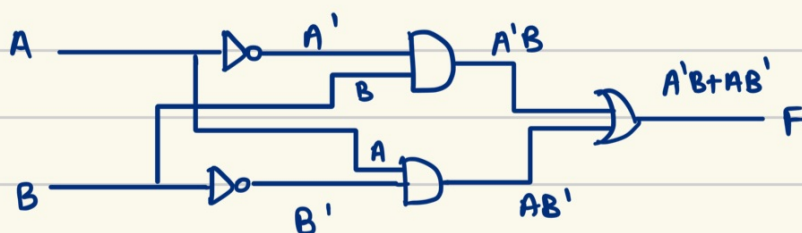
2. $F(A, B, C) = \sum m(2, 3, 4, 5)$

AB \ C	0	1
00	0	0
01	1	1
11	0	0
10	1	1

$(m_2, m_3) \rightarrow A'B$

$(m_4, m_5) \rightarrow AB'$

$\therefore F = A'B + AB'$



3. $F(A, B, C, D) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$

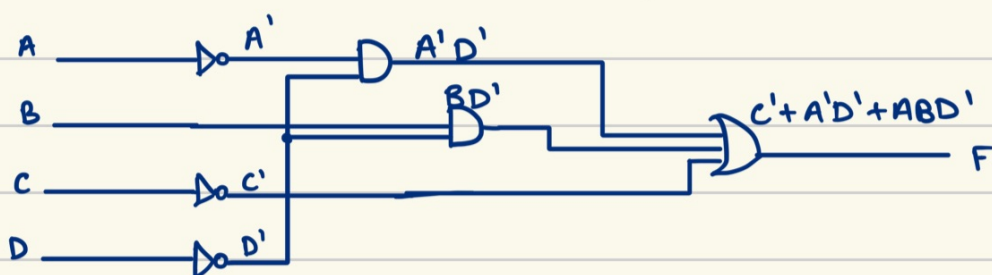
AB \ CD	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	1	1	0	1
10	1	1	0	0

$(m_0, m_1, m_4, m_5, m_{12}, m_{13}, m_8, m_9) \rightarrow C'$

$(m_0, m_4, m_2, m_6) \rightarrow A'D'$

$(m_4, m_{12}, m_6, m_{14}) \rightarrow BD'$

$\therefore F = C' + A'D' + BD'$



4. $F(A, B, C, D) = \sum m(2, 4, 6, 8, 10, 11)$

AB \ CD	00	01	11	10
00	0	0	0	1
01	1	0	0	1
11	0	0	0	0
10	1	0	1	1

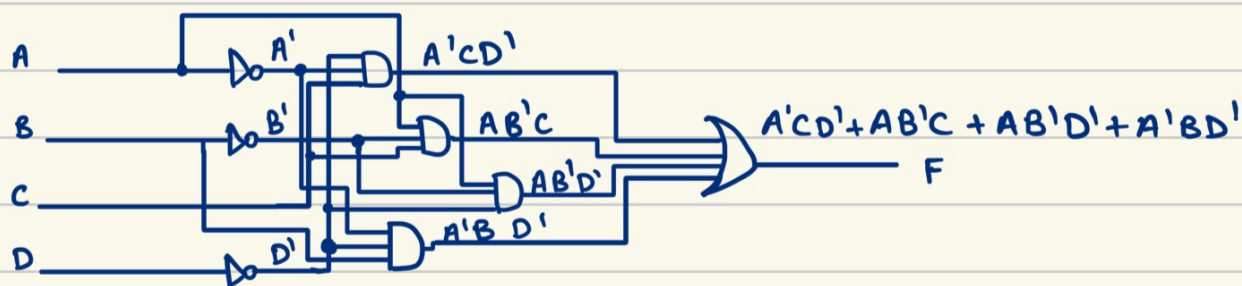
$$(m_2, m_6) \rightarrow A'CD'$$

$$(m_4, m_6) \rightarrow A'BD'$$

$$(m_{11}, m_{10}) \rightarrow AB'C$$

$$(m_8, m_{10}) \rightarrow AB'D'$$

$$\therefore F = A'CD' + A'BD' + AB'C + AB'D'$$



5. $F(A, B, C, D) = \sum m(2, 3, 4, 7)$

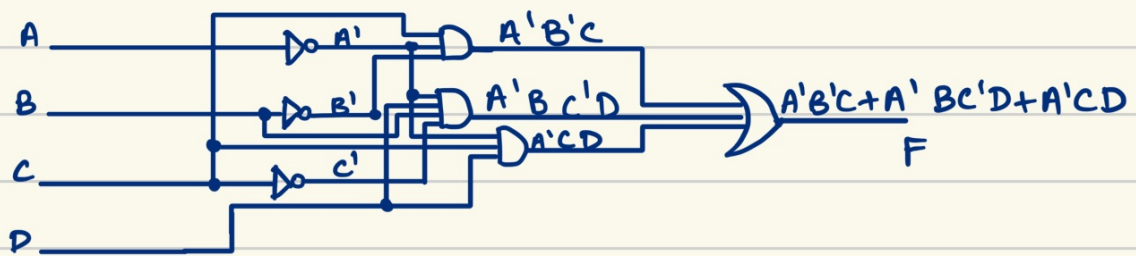
AB \ CD	00	01	11	10
00	0	0	1	1
01	1	0	1	0
11	0	0	0	0
10	0	0	0	0

$$(m_3, m_2) \rightarrow A'B'C$$

$$(m_4) \rightarrow A'BC'D'$$

$$(m_3, m_7) \rightarrow A'CD$$

$$\therefore F = A'B'C + A'CD + A'BC'D'$$



6. $F(A, B, C, D) = \prod M(2, 3, 4, 7)$

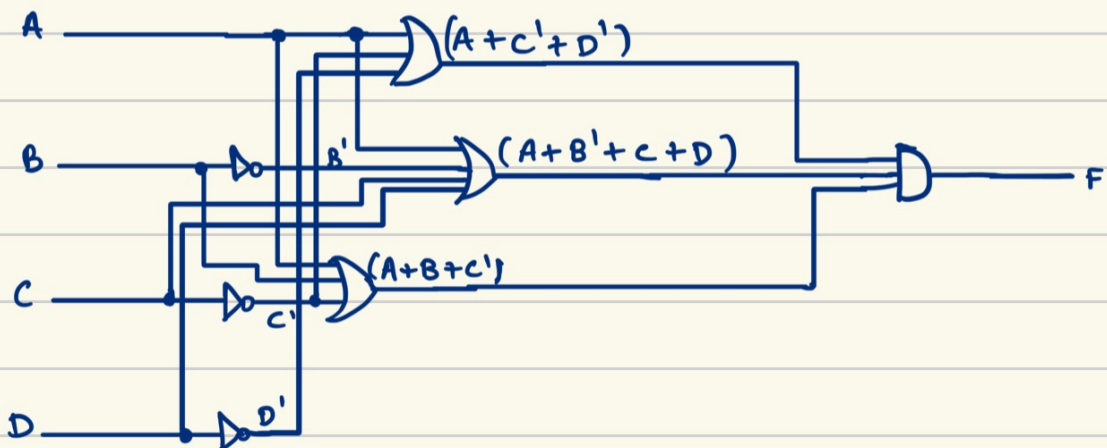
AB \ CD	00	01	11	10
00	1	1	0	0
01	0	1	0	1
11	1	1	1	1
10		1	1	1

$(m_3, m_2) \rightarrow (A + B + C')$

$(m_3, m_7) \rightarrow (A + C' + D')$

$(m_4) \rightarrow (A + B' + C + D)$

$\therefore F = (A + B + C') \cdot (A + C' + D') \cdot (A + B' + C + D)$



7. $F(A, B, C, D) = \prod M(0, 1, 2, 6, 8, 10, 12)$

AB \ CD	00	01	11	10
00	0	0	1	0
01	1	1	1	0
11	0	1	1	1
10	0	1	1	0

$$(m_0, m_1) \rightarrow (A+B+C)$$

$$(m_2, m_6) \rightarrow (A+C'+D)$$

$$(m_4, m_5) \rightarrow (A'+C+D)$$

$$(m_0, m_2, m_4, m_6) \rightarrow (B+D)$$

$$\therefore F = (A+B+C) \cdot (A+C'+D) \cdot (A'+C+D)(B+D)$$

